Notes: Gennaioli, La Porta, Lopez-de-Silanes, and Shleifer (QJE, 2013)

Human Capital and Regional Development

Contents

1 Big picture

- **Question.** What explains why some regions inside the same country are much richer than others? Is the key force geography, institutions, culture, or human capital?
- **Setting.** The paper builds a new cross-regional data set with **1,569 subnational regions in 110 countries** and combines it with firm-level data for **6,314 establishments in 20 countries** from the World Bank Enterprise Surveys.
- **Core challenge.** Cross-country regressions make it hard to separate education from institutions and other national characteristics. Even within countries, regional schooling may be endogenous because more educated people migrate toward more productive regions.
- **Main conceptual contribution.** The paper combines regional regressions, firm-level production-function evidence, and a calibrated spatial model. This lets the authors move from "education matters" to *why* it matters: through workers, entrepreneurs/managers, and possibly regional externalities.
- **Model mechanism.** The **Lucas-Lucas** model merges three ideas: (i) human capital of workers enters production directly, (ii) entrepreneurial human capital independently raises firm productivity, and (iii) regional average human capital creates spillovers. Labor is mobile across regions but housing and land are not, so migration is limited in equilibrium.

- **Main regional result.** Education dominates every other variable in explaining within-country regional income differences. In the univariate regressions, education explains far more within-country variation than geography, institutions, or culture.
- **Main multivariate result.** Once the authors include country fixed effects and standard geography controls, the coefficient on regional schooling remains large and highly significant. Measures of institutions and culture add little at the regional level.
- **Main firm-level result.** Establishments with more educated managers and more educated workers are more productive, and regional schooling also enters positively. The paper interprets this as evidence for managerial human capital and possibly regional externalities.
- **Main composition result.** Regions with a larger share of directors, officers, and college-educated managers are richer, even conditional on average education.
- **Main calibration result.** Standard development accounting understates the role of human capital because it treats returns to schooling as worker wages only. Once entrepreneurial human capital is added, the explanatory power of human capital rises sharply.
- **Economic takeaway.** The most important part of education for regional development may not be ordinary worker schooling alone, but the supply of educated entrepreneurs and managers who run firms and possibly generate local spillovers.

2 Data and measurement (key objects)

2.1 Why this paper's regional approach is useful

- The authors shift the focus from countries to **regions within countries**.
- This helps absorb country-wide institutions, legal systems, and broad cultural features using country fixed effects.
- The paper's logic is: if large income differences remain across regions inside the same country, then purely national explanations are incomplete.

Interpretation. This is the paper's key identification move. It does not claim that institutions never matter; it asks whether they explain *within-country* regional income differences once national factors are removed.

2.2 Regional data and sample construction

- Final regional sample: **1,569 regions in 110 countries**.
- Regional income data are available for **107 countries**.
- Education data are available for **106 countries and 1,519 regions**.
- The regional unit is the highest administrative level for which comparable data exist, or the nearest statistical region when administrative data are unavailable.
- The paper aggregates geography, institutions, and culture data to that regional level.

Interpretation. The sample is global in scope but the identifying variation is local. The paper is not studying a few U.S. states or European regions; it is studying almost the whole world through a comparable regional lens.

2.3 Regional income

- Regional income is measured mainly as regional GDP per capita in current PPP dollars around **2005**.
- When value-added data are unavailable, the authors use close substitutes such as income, expenditure, wages, or gross value added.
- Regional values are scaled so that population-weighted regional income adds up to national GDP in World Development Indicators.

Interpretation. The dependent variable is meant to be a harmonized regional development measure, not just a raw administrative statistic from one country.

2.4 Education

The core human-capital measure is average years of schooling of the population aged 15 and older:

$$S_i = \sum_e \big(\text{years attached to education level } e\big) \times \big(\text{population share at level } e\big). \tag{1}$$

- Primary and secondary durations come from UNESCO mappings of national school systems.
- Tertiary education is assigned four additional years.
- The paper also uses the share of the population with high-school and college degrees as alternative human-capital measures.
- A robustness check replaces current education with education of the population aged 65 and above to reduce simultaneity with current income.

Interpretation. Education is the paper's central empirical object, and the alternative measures help distinguish average schooling from the upper tail of the human-capital distribution.

2.5 Geography, institutions, and culture

The regional controls are grouped into three broad categories:

- **Geography and natural resources:** temperature, inverse distance to the coast, and oil production/reserves per capita.
- **Institutions:** an index built from Enterprise Survey variables such as informal payments, time with tax authorities, power outages, security costs, access to land, access to finance, regulatory predictability, and subnational Doing Business rankings.
- **Culture:** trust in others from the World Value Survey and the number of ethnic groups in the region.

Interpretation. These are not afterthought controls. They are the main rival explanations against which education is compared.

2.6 Firm-level data

- The firm-level sample contains up to **6,314 establishments in 20 countries and 76 regions**.
- The panel sample used for Levinsohn–Petrin estimation contains **2,922 firms in 7 countries**.
- The data contain firm sales, labor, property plant and equipment, energy use, exports, foreign ownership, years of schooling of the top manager, and years of schooling of workers.

Interpretation. The firm data let the paper look inside the regional aggregate and ask whether educated managers and workers actually show up as more productive establishments.

2.7 Occupational composition measures

To connect entrepreneurship and management more directly to development, the paper adds regional measures of workforce composition:

- share of directors and officers in the workforce,
- share of directors and officers with a college degree,
- share of employers,
- share of self-employed.

Interpretation. These variables are meant to capture whether regions with more human capital also sort more talent into entrepreneurship and management.

2.8 Descriptive facts

Table I highlights several important cross-regional inequalities:

- the average ratio of income per capita in the richest region to the poorest region within a country is **4.41**;
- the corresponding ratio for years of education is **1.80**;
- the ratio for the share with a college degree is **4.70**;
- rich and poor regions differ much more in education and entrepreneurial composition than in temperature or distance to coast.

Interpretation. The descriptive data already point toward human capital rather than geography or institutions as the most plausible account of within-country regional inequality.

3 Models and empirical specifications (all equations + interpretation)

3.1 The Lucas-Lucas regional model

The paper's theoretical structure combines entrepreneurial talent, worker human capital, and regional spillovers. At the firm level, output in region i for entrepreneur h is:

$$y_{i,h} = A_i \, h^{1-\alpha-\beta-\delta} H_{i,h}^{\alpha} K_{i,h}^{\beta} T_{i,h}^{\delta}, \qquad \alpha + \beta + \delta < 1. \tag{2}$$

Regional productivity itself depends on exogenous fundamentals and human-capital externalities:

$$A_i = \widetilde{A}_i \; E_i(h)^{\psi} L_i^{\gamma}. \tag{3}$$

Interpretation. The model separates three channels.

- Worker human capital raises productivity through the usual production channel.
- Entrepreneurial human capital enters separately and can generate very high private returns.
- Average regional human capital raises total factor productivity through spillovers.

3.2 Occupational choice inside the region

Individuals choose between entrepreneurship and wage work. In equilibrium, total human capital in a region is split between entrepreneurs and workers:

$$H_i^E = \frac{1-\alpha-\beta-\delta}{1-\beta-\delta} H_i, \qquad H_i^W = \frac{\alpha}{1-\beta-\delta} H_i. \tag{4}$$

Interpretation. The share of human capital allocated to entrepreneurship depends on how much of the private return to skill is earned as entrepreneurial profit rather than labor income.

3.3 Regional output equation

Combining production and occupational choice gives aggregate regional output:

$$Y_i = \widetilde{A}_i \, E_i(h)^{\psi} L_i^{\gamma} (H_i^E)^{1-\alpha-\beta-\delta} (H_i^W)^{\alpha} K_i^{\beta} T^{\delta}. \tag{5}$$

Interpretation. Richer regions can be rich because they have better exogenous productivity $\widetilde{A}_i$, more human capital, a more entrepreneurial workforce, stronger spillovers, or some combination of all four.

3.4 Regional income regression

After imposing capital mobility and expressing human capital in Mincerian form, the paper derives a regional estimating equation of the form:

$$\ln\left(\frac{Y_i}{L_i}\right) = C + \frac{1}{1-\beta} \ln \widetilde{A}_i + \left[1 + \frac{\gamma-\delta}{1-\beta}\right] \bar{\mu}_i S_i + \frac{\gamma-\delta}{1-\beta} \ln L_i. \tag{6}$$

Interpretation.

- The schooling coefficient combines the private productivity effect of education and any human-capital externality.
- The population coefficient captures the net effect of scale externalities minus congestion from fixed land.
- Geography, institutions, and culture enter as controls for $\widetilde{A}_i$.

3.5 Regional estimation design

The main regional regressions are OLS with **country fixed effects**. They successively add:

- geography controls,
- population,
- institutional variables,
- cultural variables,
- and alternative human-capital measures.

Interpretation. The country fixed effects are crucial. They absorb all country-wide institutions and slow-moving national characteristics, leaving only within-country regional differences to explain.

3.6 Firm-level production-function regression

At the establishment level, the paper estimates:

$$\begin{aligned}\ln y_{ij} = {} & \ln \widetilde{A}_i + (1-\alpha-\beta-\delta)\mu_{E,i}S_{E,j} + \alpha\mu_{W,i}S_{W,j} \\ & + \alpha \ln l_{ij} + \beta \ln K_{ij} + \delta \ln T_{ij} + \gamma \ln L_i + \gamma\bar{\mu}_i S_i.\end{aligned} \tag{7}$$

Empirically, the dependent variable is log gross value added (sales net of raw materials and energy), and the right-hand side includes:

- years of schooling of the manager,
- years of schooling of workers,
- regional schooling,
- log employment,
- log property plant and equipment,
- log energy expenditure,
- log population,
- geography controls,
- country and industry fixed effects.

Interpretation. The key idea is to separate *managerial schooling*, *worker schooling*, and *regional schooling*. That is what lets the paper move beyond the standard "schooling as one aggregate factor" approach.

3.7 Estimation choices at the firm level

The paper reports:

- three OLS cross-sectional specifications, and
- one Levinsohn–Petrin panel specification using energy expenditure to proxy for unobserved productivity shocks.

In the OLS cross-section, errors are clustered at the country-region level. The LP panel is used as a robustness check because inputs may be correlated with firm productivity.

Interpretation. The firm-level analysis is where the paper tries to partially break the migration and sorting problem that limits the causal interpretation of the regional regressions.

3.8 Calibration and development accounting

The paper then takes the estimated coefficients and maps them into a development-accounting exercise. It defines model-predicted income of the form:

$$\widehat{Y} = E(h)^{\gamma} L^{\gamma} H^{1-\beta-\delta} K^{\beta+\delta}, \tag{8}$$

and studies the *success ratio*

$$\text{success} = \frac{\mathrm{Var}(\log \widehat{Y})}{\mathrm{Var}(\log Y)}. \tag{9}$$

Interpretation. This measures how much of cross-country income variation the model can account for once the paper's estimates of entrepreneurial human capital and externalities are plugged in.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the regional education coefficient

The core identification strategy in the regional regressions is not an IV but a fixed-effects design:

- compare regions *within the same country*,
- control for standard geography variables,
- then ask whether education still explains regional income.

Interpretation. This is designed to shut down the usual cross-country objection that education just proxies for broader institutional quality.

4.2 Why reverse causality remains a concern

The main remaining concern is migration:

- better educated people may move toward more productive regions,
- so regional schooling may partly reflect productivity rather than cause it.

The paper is explicit that the large regional coefficient on schooling should *not* automatically be read as causal.

Interpretation. This honesty is important. The paper's contribution is not "education is cleanly identified," but rather "education dominates even after serious efforts to control for alternatives, and firm-level evidence points toward the underlying channels."

4.3 How the paper deals with omitted regional productivity

The authors try to reduce omitted-variable bias in several ways:

- they control for geography directly,
- they include country fixed effects,
- they show that institutional and cultural variables contribute very little at the regional level,
- and they run robustness checks using old-age education and regional luminosity.

Interpretation. None of these fully solves endogeneity, but together they make it harder to argue that the education coefficient is just a proxy for some obvious omitted factor.

4.4 Why firm-level evidence is useful

Firm-level regressions help because:

- they condition on firms within the same region,
- they include region and industry controls,
- and they can use panel-style methods such as Levinsohn–Petrin to handle input endogeneity.

Interpretation. If more educated managers and workers are associated with higher firm productivity inside the same region, then the regional schooling-income relationship becomes more plausible as a human-capital story.

4.5 Old-age education as a robustness check

A useful specification replaces current schooling with education of the population aged 65 and over.

Interpretation. This does not solve all endogeneity concerns, but it reduces the idea that the results are just picking up recent income-induced investments in education.

4.6 What the paper can and cannot identify

- The paper can strongly show that education is the most robust correlate of regional development within countries.
- It can show that managerial and worker schooling both matter for establishment productivity.
- It can show that entrepreneurial composition of the workforce is strongly associated with regional income.
- It cannot perfectly identify human-capital externalities because regional schooling may still be correlated with unobserved regional productivity.
- It also cannot cleanly separate entrepreneurial schooling from entrepreneurial talent.

Interpretation. The externality channel is the least secure part of the paper; the entrepreneurial human-capital channel is much more convincing.

4.7 Relation to the literature

Relative to earlier work, the paper contributes by combining:

- cross-regional development accounting,
- firm-level productivity analysis,
- occupational composition evidence,
- and a structural calibration based on a unified model.

Interpretation. The paper's real novelty is not simply "education matters," but "education matters through a richer set of channels than the standard worker-only Mincer interpretation suggests."

5 Tables and figures

5.1 Figure 1: global regional coverage

Read:

- Figure 1 maps the **1,569 regions in 110 countries** included in the sample.
- Coverage is broad across Europe, North America, Latin America, Asia, and parts of Africa.
- The map makes clear that the paper is not driven by just one country or one continent.

Economic message. The paper's evidence is unusually broad. This is a world regional-development paper, not just a cross-section of OECD subnational units.

5.2 Table 1 and Table 2: descriptive facts and univariate regressions

Read:

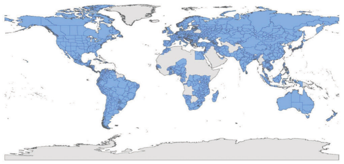

FIGURE I
Shaded Countries Are Included in Our Sample

TABLE I
DESCRIPTIVE STATISTICS

	Observations	Number of countries	Number of observations	Mean	Minimum	Maximum	Within-country range	Within-country std dev.	Average ratio region highest to lowest income per capita
			Median						
Income per capita	1,537	107	11	6,636	3,198	13,859	9,924	2,782	4.41
Years of education	1,519	107	12	6.52	5.30	8.69	2.37	0.73	1.80
Share pop with high school degree	1,525	110	12	0.18	0.12	0.25	0.11	0.03	2.45
Share pop with college degree	1,525	110	12	0.11	0.06	0.20	0.13	0.04	4.70
Population	1,569	110	12	1,284,631	330,071	3,052,762	2,458,956	873,594	3.11
Temperature	1,568	110	12	16.84	10.23	21.13	4.47	1.45	1.02
Inverse distance to coast	1,569	81	12	0.90	0.80	0.99	0.13	0.05	1.05
Oil production per capita	1,569	69	12	0.00	0.00	0.00	0.00	0.00	1.70
Institutional quality	507	110	5	−0.01	−0.09	0.10	0.15	0.07	0.12
Trust in others	745	107	9	0.23	0.12	0.38	0.22	0.07	1.25
No. ethnic groups	1,568	107	12	3.00	1.00	6.00	4.00	1.35	0.86
% Directors and officers in workforce	471	28	14	0.63	0.23	1.36	0.82	0.26	6.84
% Employers in workforce	565	35	13	3.60	2.03	5.29	2.62	0.80	2.52

Notes. The table reports descriptive statistics for the variables in the article. Columns 4 through 9 report the value of each statistic in the median country. All variables are described in Appendix B.

TABLE II
UNIVARIATE REGRESSIONS FOR REGIONAL GDP PER CAPITA

	Slope	Constant	Observations	No. countries	Within R^2	Between R^2
Years of Education	0.2866[a]	6.7245[a]	1,500	105	38%	58%
	(0.0173)	(0.1234)				
Share pop with high school degree × 12	0.3180[a]	7.7729[a]	1,506	105	15%	33%
	(0.0502)	(0.1564)				
Share pop with college degree × 16	0.2926[a]	8.1305[a]	1,506	105	27%	34%
	(0.0254)	(0.0549)				
Ln(population)	0.0536[b]	8.0211[a]	1,537	107	1%	3%
	(0.0211)	(0.2905)				
Temperature	−0.0093	8.8996[a]	1,536	107	1%	27%
	(0.0095)	(0.1418)				
Inverse distance to coast	0.8937[a]	8.0034[a]	1,537	107	4%	13%
	(0.2437)	(0.2063)				
Ln(oil production per capita)	0.1518[a]	8.7447[a]	1,537	107	2%	4%
	(0.0503)	(0.0051)				
Institutional quality	0.0801	8.5217[a]	496	79	0%	25%
	(0.3542)	(0.0000)				
Trust in others	0.0126	8.9889[a]	739	68	0%	18%
	(0.1555)	(0.0416)				
Ln(No. ethnic groups)	−0.1473[a]	8.9055[a]	1,536	107	5%	17%
	(0.0324)	(0.0322)				
% Directors and officers in workforce	0.2106[a]	8.8325[a]	447	27	15%	7%
	(0.0298)	(0.0350)				
% Employers in workforce	0.0474	8.8119[a]	553	35	3%	14%
	(0.0353)	(0.1257)				

Notes. OLS regressions of (log) regional income per capita. The independent variables are proxies for (a) education, (b) geography, (c) institutions, and (d) culture. All regressions include country dummies. The table reports the number of observations, the number of countries, the R^2 within, and the R^2 between. Robust standard errors are shown in parentheses. All variables are described in Appendix B.
[a]Significant at the 1% level. [b]Significant at the 5% level. [c]Significant at the 10% level.

- Table 1 shows that the average richest-to-poorest regional income ratio inside a country is **4.41**.
- The analogous ratio for years of education is **1.80**, and for the share with a college degree it is **4.70**.
- Table 2 shows that, in univariate country-fixed-effects regressions, education explains much more within-country regional income variation than any other variable.
- The within-country R^2 from years of education is around **38%** in Table 2, whereas no other single variable exceeds single-digit explanatory power in a comparable way.

Economic message. Before controls, before theory, and before calibration, the raw descriptive picture already says that education is the dominant regional correlate.

5.3 Figure 2 and Figure 3 plus Table 4: the regional education-income relationship

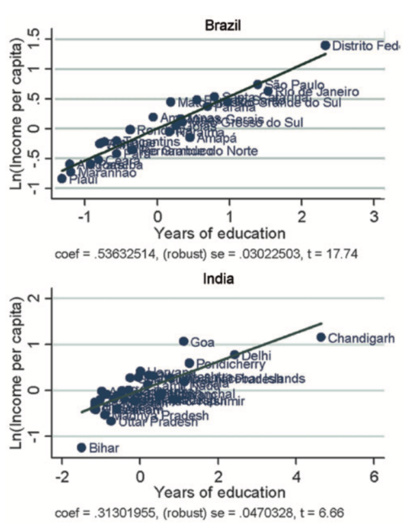

FIGURE II
Partial Correlation Plot of (log) Regional Income per Capita and Years of Education in Brazil and India

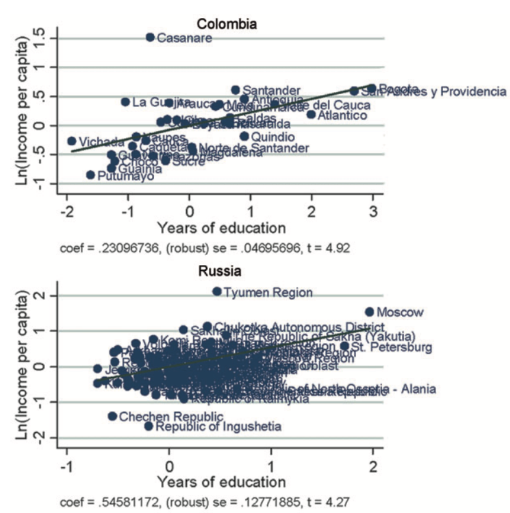

FIGURE II (Continued)
Partial Correlation Plot of (log) Regional Income per Capita and Years of Education in Colombia and Russian Federation

Read:

- Figure 2 shows strong partial correlations between education and regional income in Brazil, India, Colombia, and Russia.
- Figure 3 shows the same relationship using the full global sample after controlling for temperature, inverse distance to coast, oil, population, and country dummies.
- In Table 4, the coefficient on years of education is about **0.28** in the baseline regional specification with geography and population controls.
- When institutions and culture are added, the education coefficient remains highly significant and rises to about **0.37** in the most comprehensive specification.
- Replacing current education with education of the population aged 65+ still yields a coefficient of about **0.25**.

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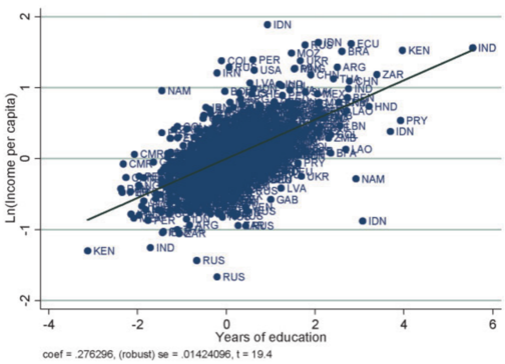

FIGURE III

Partial Correlation Plot of (log) Regional Income per Capita and Years of Education Controlling for Temperature, Inverse Distance to Coast, Ln(oil production per capita), Ln(population), and Country Dummies

TABLE IV

REGIONAL INCOME PER CAPITA, GEOGRAPHY, INSTITUTIONS, AND CULTURE

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Temperature	−0.0156[c]	−0.0128	−0.0069	0.0003	−0.0142	0.0020	−0.0095[c]
	(0.0082)	(0.0083)	(0.0053)	(0.0063)	(0.0089)	(0.0081)	(0.0047)
Inverse distance to coast	1.0283[a]	0.5236[a]	0.5066	0.5806[b]	0.4568[a]	0.5713[c]	0.8842[a]
	(0.2080)	(0.1380)	(0.3257)	(0.2377)	(0.1292)	(0.3397)	(0.2547)
Ln(oil production per capita)	0.1650[a]	0.1848[a]	0.1604	0.1459[b]	0.1983[a]	0.1041	0.1403[b]
	(0.0477)	(0.0470)	(0.0970)	(0.0593)	(0.0491)	(0.2006)	(0.0643)
Years of education		0.2763[a]	0.3476[a]	0.3032[a]	0.2653[a]	0.3678[a]	
		(0.0170)	(0.0215)	(0.0278)	(0.0178)	(0.0443)	
Ln(population)		0.0122	0.0008	0.0091	0.0165	0.0050	−0.0259
		(0.0164)	(0.0215)	(0.0177)	(0.0169)	(0.0393)	(0.0177)
Institutional quality			0.3667			0.4667	
			(0.2297)			(0.2850)	
Trust in others				−0.0413		0.0439	
				(0.0879)		(0.1632)	
Ln(no. ethnic groups)					−0.0499[b]	0.0005	
					(0.0243)	(0.0490)	
Years of education 65+							0.2515[a]
							(0.0283)
Constant	8.1061[a]	6.3594[a]	5.9375[a]	5.9902[a]	6.5044[a]	5.4934[a]	7.7483[a]
	(0.2277)	(0.1857)	(0.4235)	(0.2809)	(0.1637)	(0.6989)	(0.2680)
Observations	1,536	1,499	483	728	1,498	281	608
Number of countries	107	105	78	66	105	45	39
Within R^2	8%	42%	62%	48%	42%	62%	39%
Between R^2	47%	60%	61%	51%	60%	51%	62%
Within R^2 excluding institutions and culture	8%	42%	61%	48%	42%	61%	39%
Within R^2 excluding education	8%	10%	6%	12%	15%	16%	9%
Between R^2 excluding institutions and culture	47%	60%	60%	51%	60%	50%	62%
Between R^2 excluding education	48%	42%	46%	6%	47%	63%	68%

Notes. Country fixed-effects regressions of (log) regional income per capita. All regressions include temperature, inverse distance to coast, and (log) per capita oil production and reserves. In addition, regressions include measures of (a) human capital, (b) geography, (c) institutions, and (d) culture. Robust standard errors are shown in parentheses. For comparison, the bottom panel shows the adjusted R^2 of two alternative specifications: (a) a regression that excludes the relevant measure of institutions or culture; and (b) a regression that excludes education. All variables are described in Appendix B.

[a]Significant at the 1% level. [b]Significant at the 5% level. [c]Significant at the 10% level.

The final regression in Table IV addresses the concern that :he coefficient on education is biased because richer regions

Economic message. The education result is not fragile. It survives geography controls, institutional and cultural controls, and a robustness check designed to reduce contemporaneous simultaneity.

5.4 Table 5: firm-level productivity and schooling

Read:

- Table 5 is the core firm-level production-function table.
- In OLS, the coefficient on manager schooling ranges from about **0.015 to 0.026**, while the coefficient on worker schooling ranges from about **0.015 to 0.017**.
- Regional schooling also enters positively, around **0.07 to 0.09** in the OLS specifications.
- Log population is positive, roughly **0.10 to 0.12**, consistent with agglomeration or scale effects.

- In the Levinsohn–Petrin specification, labor is about **0.62**, property plant and equipment about **0.34**, manager schooling about **0.026**, and worker schooling about **0.026**. Regional schooling becomes imprecise because the panel is much smaller.

Economic message. Educated managers matter a lot, educated workers matter too, and the data are consistent with human capital operating through multiple channels rather than one aggregate wage-based channel.

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TABLE V
GROSS VALUE ADDED

	OLS (1)	OLS (2)	OLS (3)	Levinsohn-Petrin (4)
Temperature	0.0505^{b}	0.0251	0.0303^{c}	0.0698^{a}
	(0.0226)	(0.0183)	(0.0180)	(0.0197)
Inverse distance to coast	−0.1979	−0.2579	−0.3264	−0.2429
	(0.4519)	(0.4748)	(0.5051)	(0.5333)
Ln(oil production per capita)	−1.4113^{c}	−1.1546	−1.1133	15.4289
	(0.7138)	(0.7858)	(0.8374)	(45.4751)
Years of education	0.0730^{a}	0.0765^{a}	0.0866^{a}	−0.0087
	(0.0228)	(0.0200)	(0.0207)	(0.0317)
Ln(population)	0.1263^{b}	0.0967^{b}	0.1010^{b}	0.0135
	(0.0481)	(0.0445)	(0.0464)	(0.0938)
Years of education of manager	0.0263^{a}	0.0164^{a}	0.0147^{a}	0.0256^{a}
	(0.0052)	(0.0049)	(0.0049)	(0.0090)
Years of education of workers	0.0169^{b}	0.0149^{c}	0.0146^{c}	0.0265^{a}
	(0.0078)	(0.0076)	(0.0075)	(0.0100)
Ln(no. employees)	0.8602^{a}	0.6757^{a}	0.6399^{a}	0.6151^{a}
	(0.0340)	(0.0279)	(0.0265)	(0.0301)
Ln(property, plant, and equipment)	0.2434^{a}	0.1668^{a}	0.1614^{a}	0.3450^{a}
	(0.0169)	(0.0164)	(0.0161)	(0.0493)
Ln(expenditure on energy)		0.2548^{a}	0.2457^{a}	
		(0.0227)	(0.0227)	
Ln(1 + firm age)			0.0348^{c}	−0.0325
			(0.0182)	(0.0286)
Multiple establishments			0.1522^{a}	
			(0.0377)	
% Export			0.0017^{a}	
			(0.0006)	
% Equity owned by foreigners			0.0032^{a}	
			(0.0006)	
Constant	2.1234^{b}	2.6136^{a}	2.5454^{a}	
	(0.9712)	(0.9128)	(0.9378)	
Observations	6,314	6,314	6,312	2,922
Number of countries	20	20	20	7
Within R^2	73%	75%	76%	
Between R^2	35%	78%	76%	

Notes. The table reports regressions for (log) sales minus expenditure on raw materials and energy. The first three columns show fixed-effect regressions for the cross-section, while the last column shows Levinsohn and Petrin (2003) panel regressions. All regressions include country and industry fixed effects. The errors of the fixed-effect regression are clustered at the country-regional level. Robust standard errors

TABLE VI
REGIONAL INCOME PER CAPITA AND THE COMPOSITION OF HUMAN CAPITAL

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Temperature	−0.0136^{c}	−0.0127	−0.0097	−0.0063	−0.0087	−0.0048	−0.0112^{b}	−0.0070
	(0.0078)	(0.0084)	(0.0073)	(0.0061)	(0.0063)	(0.0054)	(0.0055)	(0.0051)
Inverse distance to coast	0.6476^{a}	0.5207^{a}	0.6659^{b}	0.5047^{b}	0.6665^{b}	0.5138^{b}	0.4965^{b}	0.3618^{c}
	(0.1603)	(0.1395)	(0.2593)	(0.2417)	(0.2566)	(0.2382)	(0.1990)	(0.1818)
Ln(oil production per capita)	0.1808^{a}	0.1881^{a}	0.1081^{c}	0.1132^{b}	0.1120^{b}	0.1186^{b}	0.1405^{b}	0.1426^{b}
	(0.0463)	(0.0463)	(0.0560)	(0.0533)	(0.0528)	(0.0502)	(0.0546)	(0.0532)
Share Pop with high school degree × 12	0.2024^{a}		0.2408^{a}	−0.1089	0.2427^{a}	−0.1020^{c}	0.1829^{a}	−0.1121^{c}
	(0.0309)		(0.0570)	(0.0652)	(0.0552)	(0.0579)	(0.0550)	(0.0619)
Share Pop with college degree × 16	0.2488^{a}	0.0835^{a}	0.2350^{a}	−0.0175	0.2323^{a}	−0.0175	0.2343^{a}	0.0171
	(0.0210)	(0.0256)	(0.0398)	(0.0439)	(0.0401)	(0.0412)	(0.0344)	(0.0348)
Years of education		0.2246^{a}		0.3806^{a}		0.3708^{a}		0.3249^{a}
		(0.0245)		(0.0677)		(0.0624)		(0.0471)
Ln(population)		0.0045		−0.0255^{b}		−0.0237^{b}		−0.0233^{c}
		(0.0160)		(0.0115)		(0.0098)		(0.0117)
% Directors and officers in workforce			0.0839^{a}	0.0725^{a}				
			(0.0208)	(0.0199)				
% Directors and officers with a college degree					0.1266^{b}	0.1117^{a}		
					(0.0505)	(0.0368)		
% Employers in workforce							0.0284^{c}	0.0184
							(0.0153)	(0.0141)
% Self-employed in workforce							−0.0148^{a}	−0.0135^{a}
							(0.0044)	(0.0037)
Constant	7.2338^{a}	6.6545^{a}	7.3371^{a}	6.6233^{a}	7.3566^{a}	6.6860^{a}	7.9530^{a}	7.3628^{a}
	(0.2321)	(0.1990)	(0.2495)	(0.3063)	(0.2418)	(0.2803)	(0.2733)	(0.2938)
Observations	1,505	1,499	446	441	476	471	551	546
Number of countries	105	105	27	27	28	28	35	35
Within R^2	39%	43%	49%	58%	48%	56%	49%	56%
Between R^2	54%	61%	64%	83%	63%	82%	76%	84%

Notes. Country fixed effects regressions of (log) regional income per capita. Robust standard errors are shown in parentheses. The table reports the number of observations, the number of countries, the R^2 within, and the R^2 between. All variables are described in Appendix B.
[a]Significant at the 1% level. [b]Significant at the 5% level. [c]Significant at the 10% level.

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5.5 Table 6: the composition of human capital

Read:

- Table 6 asks whether *who* in the region is educated matters, not just average schooling.
- A higher share of college-educated population is more strongly related to income than a higher share with only high school.
- A 1 percentage point increase in the share of directors and officers in the workforce is associated with roughly an **8%** increase in GDP per capita.
- A 1 percentage point increase in the share of college-educated directors and officers is associated with an increase in GDP per capita of roughly **11–12%**.
- The employer share has a smaller effect and becomes insignificant once education is controlled for more tightly.

TABLE VII

CALIBRATION EXERCISE

Panel A: $(1 - (\alpha+\beta+\delta)) * \mu_E = 0.030$ and $\alpha * \mu_W = 0.030$											
α	50.0%	51.0%	52.0%	53.0%	54.0%	55.0%	56.0%	57.0%	58.0%	59.0%	60.0%
μ_E	20%	21%	23%	25%	27%	30%	33%	38%	43%	50%	60%
μ_W	6.0%	5.9%	5.8%	5.7%	5.6%	5.5%	5.4%	5.3%	5.2%	5.1%	5.0%
μ_{avg}	11%	12%	13%	15%	18%	21%	26%	33%	42%	55%	74%
$\frac{\text{Wage entrepreneur}}{\text{wage worker}}$	10.8	13.3	16.9	22.3	30.9	45.5	73.1	131.8	280.9	768.2	3133.8
Without externalities											
$\sigma^2_{\hat{Y}}$	0.83	0.85	0.87	0.93	0.98	1.10	1.23	1.48	1.84	2.46	3.57
σ^2_{Y}	1.32	1.32	1.32	1.32	1.32	1.32	1.32	1.32	1.32	1.32	1.32
$\frac{\sigma^2_{\hat{Y}}}{\sigma^2_{Y}}$	62%	64%	66%	70%	74%	83%	93%	112%	139%	186%	269%
With externalities											
$\sigma^2_{\hat{Y}}$	0.91	0.95	1.00	1.09	1.20	1.42	1.67	2.17	2.93	4.24	6.63
σ^2_{Y}	1.32	1.32	1.32	1.32	1.32	1.32	1.32	1.32	1.32	1.32	1.32
$\frac{\sigma^2_{\hat{Y}}}{\sigma^2_{Y}}$	69%	72%	75%	83%	91%	108%	127%	164%	221%	320%	501%

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Economic message. The data fit the model's entrepreneurial channel very well. It is not just education in the abstract; it is education sorted into entrepreneurship and management.

5.6 Table 7: calibration and development accounting

Read:

- Under a standard development-accounting benchmark with an 8% Mincerian return and no externalities, physical and human capital explain about **57%** of the variance in income per worker.
- When the average return is raised to **20%** to reflect the paper's entrepreneurial-human-capital estimates, the success ratio rises to about **81%**.
- Table 7 shows that as the labor share rises, implied entrepreneurial returns become very large, while worker returns remain modest.
- In the paper's benchmark range, externalities matter, but mainly when entrepreneurial returns are already high.

Economic message. The calibration turns the firm-level estimates into a sharp macro conclusion: standard worker-only returns to schooling are too small to account for development differences, but worker plus entrepreneurial schooling can account for much more.

6 Short summary

- The paper studies regional development inside countries using a huge cross-regional data set and complementary firm-level data.
- Education is by far the strongest regional correlate of income within countries; institutions and culture contribute little at that level.
- The authors build a Lucas-Lucas model in which worker schooling, entrepreneurial schooling, and regional spillovers all matter.
- Firm-level regressions show that educated managers and workers are both associated with higher productivity, and regional schooling also enters positively.

- Regional occupational composition matters too: regions with more directors, officers, and college-educated managers are richer.
- Calibration implies that development accounting understates the contribution of human capital if it ignores entrepreneurial returns.
- The paper's broad conclusion is that raising education matters not only because it creates better workers but also because it creates better entrepreneurs and managers, and perhaps stronger regional spillovers.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that human capital, measured through education, is the single most important observable variable for explaining regional income differences within countries. It further shows that this relationship likely operates through several distinct channels: worker education, entrepreneurial or managerial education, and possibly externalities tied to the quality of local human capital.

7.2 Economic intuition

A region becomes richer not only because educated workers are more productive on the job, but also because educated people are more likely to become effective entrepreneurs and managers who organize production, run better firms, and potentially create knowledge spillovers for others nearby. Once that channel is allowed for, the role of education in development becomes much larger than in standard worker-only accounting exercises.

7.3 Takeaway

The main lesson is that development accounting should not stop at worker Mincer returns. Human capital matters much more broadly. Regions prosper when they accumulate educated people who both work productively and sort into entrepreneurship and management, and those channels appear far more powerful than regional institutions or culture in this paper's within-country evidence.